

Borrower Health Index and Recovery Estimation

Reconstructing a Credit Bureau Score from Repayment Behaviour

1 Introduction

The Home Credit Default Risk dataset includes `EXT_SOURCE_1/2/3`: proprietary external credit scores derived from credit bureau records spanning a borrower’s complete history across all lenders. These features rank among the strongest predictors available in the dataset.

A bureau score is comprehensive in breadth. A lender’s own installment records are narrower in scope but substantially denser in temporal granularity, capturing exact payment timing, partial settlement behaviour, and prioritisation between competing obligations.

This project quantifies the trade-off between these information sets:

How much of a commercial bureau score can be reconstructed from repayment behaviour alone?

The pipeline is constructed as an internal, behaviour-only credit score and evaluated against Home Credit’s external score on identical held-out borrowers.

Previously. External scores treated as ordinary input features, entering the model alongside behavioural features.

Issue. With the strongest available predictor supplied directly as an input, the pipeline’s own contribution could not be isolated. Any performance figure would conflate signal extracted from repayment behaviour with signal supplied by the bureau, leaving the stated question unanswerable.

Revised. External scores held out as an independent benchmark and scored separately, so the behavioural pipeline is measured against them rather than through them. An ablation quantifying their standalone contribution is reported in Section 6.1.

2 Data

Table 1: Data sources and their contribution to the modelling panel

Source	Contribution
<code>installments_payments.csv</code>	Per-installment repayment behaviour: amount due, amount paid, payment timing
<code>application_train.csv</code>	Borrower context: income, credit exposure, employment tenure, <code>EXT_SOURCE_*</code> , <code>TARGET</code>

16,632 borrowers loaded; 12,991 retained following the baseline estimation window; 1,673 reach final evaluation, of which 138 are eventual defaulters.

Previously. Sample of 3,000 borrowers, with only **TARGET** drawn from the application table.

Issue. The sample yielded too few defaulters for stable interval estimation, and the application table — already being read — was contributing a single label column while its borrower-context fields were discarded. Income, credit exposure and employment tenure were available at no additional cost and were not being used.

Revised. Sample increased to 20,000 borrowers, and ten borrower-context features derived from the application table, including credit-to-income and annuity-to-income ratios, employment tenure and regional population density.

Previously. Loader falling back to a synthetic panel when the source files were absent.

Issue. A silent fallback permits results to be produced from generated data without any indication in the output, and any figure so produced is indistinguishable from a genuine one downstream.

Revised. The loader raises on missing source files. No synthetic path exists in the pipeline.

Previously. Baseline estimation window of three months.

Issue. Borrowers average approximately six usable observation periods. Reserving three for baseline estimation left too few for trajectory construction, and the loss compounded with the lagging requirement, which consumes a further period.

Revised. Window reduced to two periods, retaining sufficient data for stable stress estimation while releasing an additional period to the modelling panel.

Temporal representation. Home Credit provides day-offsets rather than calendar dates. Monthly periods are approximated as 30.44-day buckets measured from each borrower’s first observed installment. The anchoring choice is material and is discussed in Section 5.6.

3 Feature Construction

The behavioural score derives exclusively from repayment records, placing the burden of representation on feature construction.

3.1 Signed payment timing

PAYMENT_WINDOW_DAYS preserves sign, with negative values indicating early settlement.

Previously. Payment timing encoded as **DAYS_PAST_DUE**, clipped at zero.

Issue. Clipping discards the distinction between borrowers settling ahead of schedule and those settling precisely on time. These are behaviourally distinct populations with different risk profiles, collapsed into a single value of zero.

Revised. Signed timing retained as a separate feature, with the clipped variant kept alongside it for the “how overdue” signal specifically.

PAY_WINDOW_STD_3M captures dispersion in payment timing independently of its central tendency.

Previously. Rolling mean of payment timing (**PAY_WINDOW_ROLLING_3M**) and its deviation from the current value (**TIMING_DRIFT**).

Issue. A mean cannot distinguish a borrower consistently five days late, reflecting a stable payday cycle, from one alternating between ten days early and fifteen days late, reflecting irregular income. Both produce identical rolling means.

Revised. Rolling standard deviation added, capturing dispersion separately from central tendency.

3.2 Payment fragmentation

The installment data records multiple payment rows against a single due installment, representing an obligation settled across several transactions.

Previously. Each payment row treated as an independent observation.

Issue. A EUR 400 obligation settled as three transactions of EUR 150, EUR 150 and EUR 100 was indistinguishable from a single EUR 400 settlement. The obligation is met in full and on schedule in both cases, but the fragmented case indicates a liquidity constraint.

Revised. `N_PAYMENT_SPLITS` counts payment rows per due installment; `SPLIT_PAYMENT_RATE_3M` tracks the rolling share of obligations requiring multiple transactions.

This signal exists only in transaction-level records held by the originating lender and is not observable at bureau level.

3.3 Obligation prioritisation

Borrowers holding multiple obligations do not default uniformly. Under liquidity pressure they triage, protecting certain obligations while permitting others to lapse. A Gaussian mixture model over obligation-level features infers which obligations are prioritised (Section 4.3). `PRIORITY_SPREAD` measures divergence in inferred priority within a borrower-month, and `SELECTIVE_FAILURE` measures the corresponding divergence in realised payment performance between the highest- and lowest-priority obligations.

3.4 Trend features

`PAYMENT_RATIO_TREND_3M` and `DPD_TREND_3M` encode direction of movement rather than level, constructed entirely from lagged observations.

Previously. Direction inferred implicitly from the difference between a current value and its rolling mean.

Issue. A deviation from a rolling mean conflates two distinct situations: a single anomalous period against a stable history, and a sustained trend that the rolling window has partly absorbed. The two carry different risk implications and were not separable.

Revised. Explicit trend features differencing lagged values across a fixed interval, so directional movement is represented independently of deviation from a moving average.

4 Methodology

4.1 Framing

Conventional credit modelling treats default as binary classification over a borrower snapshot. This project adopts a framing drawn from industrial reliability engineering, in which assets are assigned a continuously measured condition and Remaining Useful Life models forecast the degradation trajectory to enable intervention prior to failure.

Table 2: Correspondence between industrial asset monitoring and consumer credit

Industrial asset monitoring	Consumer credit
Measured asset condition	Health Score $H(t)$
Degradation rate	Forecast $\Delta H(t+1)$
Remaining Useful Life	Time to default
Maintenance scheduling	Collections prioritisation
Cost of unplanned failure	Unrecovered balance

Previously. Default modelled as binary classification over a borrower snapshot.

Issue. A binary formulation produces a decision but no timing. It cannot express how far a borrower is from the threshold, in which direction they are moving, or how much time remains before intervention becomes necessary — all of which are operationally required for collections prioritisation.

Revised. Continuous health score with an explicit degradation trajectory, yielding alert timing, monitoring tier assignment and expected recovery alongside the binary decision.

4.2 Baseline stress estimation

Each borrower is assigned a latent financial stress parameter θ_i , with obligation-level priority p_{ij} , under the generative model

$$r_{ij} = 1 - \theta_i(1 - p_{ij}) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2) \quad (1)$$

$$\theta_i \sim \text{TruncNormal}(0, \tau_\theta^2) \text{ on } [0, \infty) \quad (2)$$

$$p_{ij} \sim \text{TruncNormal}(0.5, \tau_p^2) \text{ on } [0, 1] \quad (3)$$

where r_{ij} is the realised payment ratio. A borrower under stress underpays, except on obligations treated as high priority. Both conditional distributions are linear-Gaussian given the other parameter, so each Gibbs update reduces to a closed-form truncated-normal draw. The sampler runs 300 iterations with 100 discarded to burn-in.

Previously. θ_i estimated over each borrower’s complete observation history.

Issue. An observation at month 5 carried a parameter informed by months 6 onward. Because θ_i appears static, this transmitted future information through a variable that does not visibly vary. This defect alone accounted for a substantial share of an implausible reported $R^2 = 0.985$.

Revised. θ_i estimated exclusively from each borrower’s initial observation window, then frozen. Window observations are excluded from the modelling panel, so no row is scored using a parameter it helped estimate.

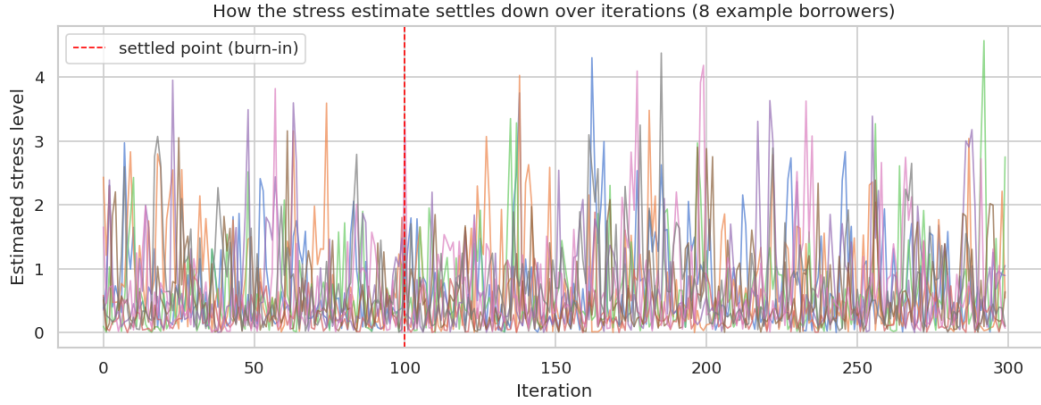


Figure 1: Gibbs sampler traces for eight randomly selected borrowers. The vertical rule marks the burn-in cutoff at 100 iterations; chains are stationary well before this point. Posterior means over the retained draws form the frozen θ_i estimates.

4.3 Obligation prioritisation model

Why a mixture model

When a borrower holds several concurrent obligations, the quantity of interest is not observable: no field records which loan the borrower is choosing to protect. What is observable is the outcome of that choice — some obligations settle promptly and in full, others lapse, within the same borrower and the same month.

This is a latent class problem. The population of obligations is assumed to arise from two unobserved regimes, one prioritised and one deprioritised, and the task is to recover regime membership from observed settlement behaviour. A Gaussian mixture is the natural estimator for this structure: it models the observed distribution as arising from unobserved components, and returns a posterior probability of membership rather than a hard assignment.

The soft assignment matters downstream. `PRIORITY_SPREAD` measures the divergence in inferred priority *within* a borrower-month, which requires a continuous membership score; a hard clustering would collapse it to a binary indicator and discard the degree of divergence, which is precisely the stress signal of interest.

Two components were specified rather than selected by information criterion. The construct being estimated is binary by definition — an obligation is either being protected or it is not — so component count is fixed by the modelling question rather than by fit statistics.

Specification

A two-component Gaussian mixture over eleven obligation-level features. The reliable component is identified by lowest mean days-past-due.

Previously. Five features: installment amount, days past due, signed payment window, payment count, and baseline stress.

Issue. The model was assigning priority from a narrow view of each obligation, with no access to the recent behavioural context already engineered elsewhere in the pipeline. Priority inference was consequently coarser than the information available.

Revised. Expanded to eleven features, adding lagged payment ratio, lagged and rolling maximum days-past-due, payment fragmentation, and timing dispersion — giving the prioritisation model the same behavioural depth as the forecasting stage.

Previously. Component identified by fixed positional index into the fitted means array.

Issue. The index silently referred to a different feature whenever the feature list changed, selecting the wrong component without raising an error.

Revised. Component resolved by feature name, so the lookup remains correct under feature list modification.

Previously. Priority aggregated as a within-month mean (`AVG_PRIORITY_SCORE`), recomputed independently each period.

Issue. A memoryless per-period average discards continuity and admits period-to-period noise directly into the health scorecard.

Revised. `CURRENT_PRIORITY` maintained as an exponentially-weighted running estimate, so each period's mixture output refines a carried-forward prior rather than replacing it.

The mixture model is fit exclusively on training borrowers; test observations are scored by transform without refitting.

4.4 Health scorecard

The health score is a logistic regression estimating probability of good standing, consistent with standard credit scorecard construction and yielding directly interpretable per-feature coefficients.

Previously. A second Gaussian mixture clustering borrower-months into latent health regimes, with the health score derived from component posterior probabilities.

Issue. Clustering is unsupervised, whereas a credit score is a supervised risk estimate. The posterior probabilities additionally concentrated at the boundaries of $[0, 1]$, producing a bimodal distribution that destabilised downstream trajectory forecasting.

Revised. Logistic regression against realised outcome, producing a smooth score with signed, interpretable coefficients.

Previously. Fixed default regularisation.

Issue. Four correlated delinquency features assumed large opposing coefficients, causing `MAX_DPD` to acquire a positive sign — implying that greater delinquency implies improved health. A multicollinearity artifact rather than a substantive effect.

Revised. Regularisation strength selected by five-fold cross-validation, shrinking correlated coefficients toward one another and restoring coefficient coherence.

Previously. Temporal smoothing by centered median window over $(t-1, t, t+1)$.

Issue. A centered window smooths period t using period $t+1$, transmitting forward information into a feature intended for causal use.

Revised. Trailing median window over $(t-2, t-1, t)$.

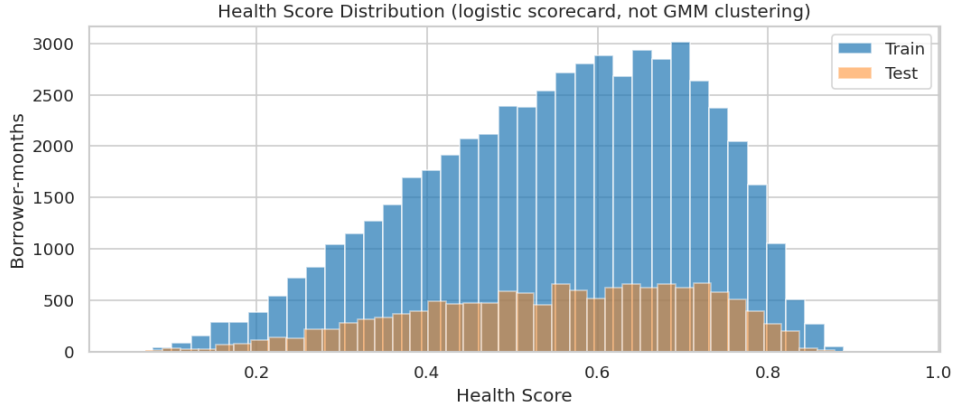


Figure 2: Distribution of the smoothed health score across borrower-months, train and test. The distribution is unimodal and continuous, in contrast to the boundary concentration produced by the superseded mixture-model formulation.

4.5 Trajectory forecasting

The forecast target is $\Delta \text{logit}(H)$ rather than ΔH .

Previously. Forecasting ΔH directly in probability space, with the result clipped to $[0, 1]$ after each step.

Issue. Since H is bounded, an additive step carries position-dependent meaning: a movement of 0.05 is routine at $H = 0.50$ and infeasible at $H = 0.02$. Under recursive simulation, clipping caused trajectories to saturate at the boundaries and remain there, producing a degenerate bimodal outcome distribution.

Revised. Forecasting in logit space, which is unbounded and renders step size scale-invariant. The inverse sigmoid transform enforces the $[0, 1]$ bound by construction, approaching the boundaries asymptotically rather than by truncation.

Defaulter observations are upweighted by inverse frequency at approximately $12\times$. All features are lagged to the prior period (Section 5.4). Feature importance is well distributed, with the leading feature accounting for 6.5% of total gain.

4.6 Recursive simulation

Forward simulation feeds each prediction into the subsequent step over a 36-month horizon.

Previously. Trajectory advanced by a fixed parametric decay function.

Issue. The fitted model was not consulted during simulation. The trajectory followed a hard-coded functional form regardless of borrower state, so the forecast reflected the decay equation rather than the model.

Revised. The fitted model is queried at every recursive step, with all lag features updated between steps so that each prediction conditions on the simulated state.

Five controls govern multi-step behaviour:

Velocity clipping

constrains each prediction to the empirical training range.

Momentum blending

combines each prediction with prior-period velocity under time-decay weighting, preventing any single prediction from dominating the trajectory.

Feature capping

at the 99th training percentile constrains the simulation to the model’s observed input space.

Phase scaling

applies lenient adjustment during the initial observation period, when limited confirmed history is available, and stricter adjustment once behaviour is established.

Persistence gating

requires a directional shift to repeat across consecutive periods before amplification is applied.

Previously. Blending weight made responsive to the magnitude of deviation between each new prediction and established momentum, with the result amplified.

Issue. The intent was to react quickly to emerging deterioration. In practice, single-step deviations are dominated by noise, so the mechanism amplified noise uniformly across the population. Alert rates rose for defaulters and non-defaulters alike and discrimination degraded.

Revised. Blending weight reverted to time-decay. Amplification retained but gated on persistence, requiring a direction to repeat across consecutive periods before it is treated as a pattern.

The simulation produces trajectory forecasts, alert timing with quantified lead time, and monitoring tier assignment.

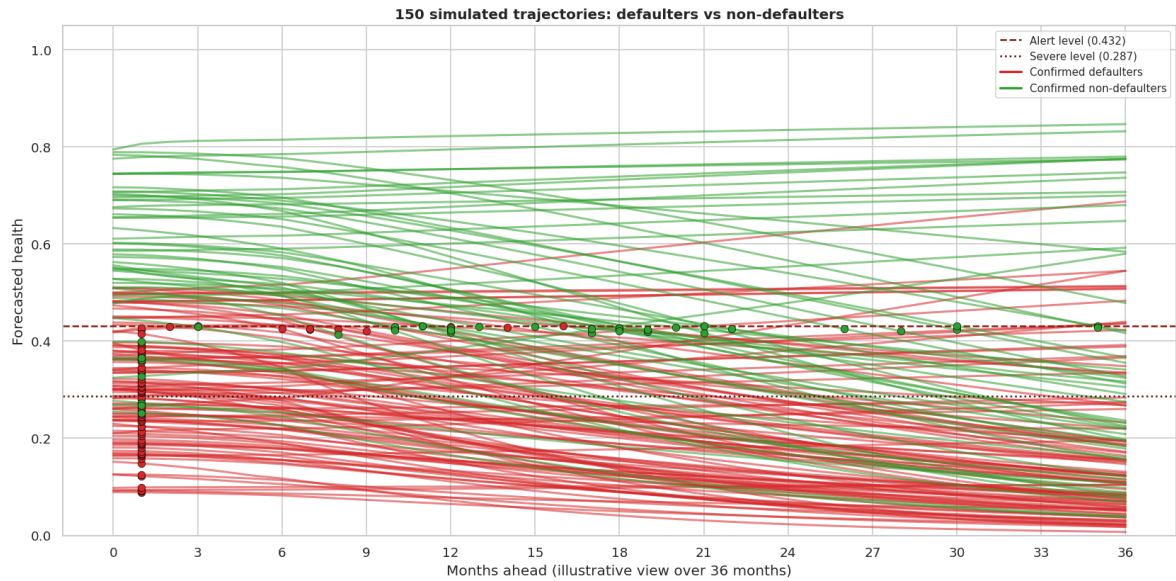


Figure 3: Simulated health trajectories over a 36-month horizon for 150 test borrowers, coloured by realised outcome. Markers indicate the period at which each trajectory first crosses the alert threshold. Separation between outcome groups is concentrated in the early forecast periods, motivating the shortened calibration horizon (Section 4.8).

4.7 Expected recovery

An isotonic regression maps forecasted health to expected payment ratio under a monotonicity constraint, ensuring that higher forecast health cannot imply lower expected payment. Fit on training data only and multiplied by the outstanding installment amount, this yields a per-account expected shortfall supporting collections prioritisation prior to the payment date.

Previously. Unconstrained regression mapping forecast health to expected payment ratio.

Issue. An unconstrained fit admits non-monotone regions in which a higher forecast health implies a lower expected payment. Such regions are artifacts of local sample density rather than substantive effects, and they are indefensible in a system whose output informs collections decisions.

Revised. Isotonic regression, which enforces monotonicity by construction rather than recovering it incidentally from the data.

4.8 Threshold calibration

The system must convert a continuous health forecast into a binary alert decision. The calibration question is what value that threshold should take, and against what the threshold should be scored. Three formulations were evaluated.

Formulation I: population percentile

Previously. Alert threshold set at the 30th percentile of the simulated health distribution, severe threshold at the 10th.

Issue. A population percentile mechanically fixes the alert rate. Since non-defaulters constitute approximately 92% of the population and therefore dominate the distribution, “bottom 30% of the population” is approximately “30% of non-defaulters” by construction. The false positive rate was consequently pinned near 30% irrespective of any improvement to the underlying model — the constraint was imposed by the threshold definition rather than by model performance.

Revised. Superseded by Formulation III.

This formulation also produced degenerate thresholds under certain configurations: where the simulated distribution carried a point mass at a boundary, the 10th and 30th percentiles resolved to the same value and the two alert tiers collapsed into one.

Formulation II: calibrated default probability

Previously. $P(\text{default} \mid H)$ estimated by isotonic regression on realised outcomes, with the threshold set where predicted probability reaches a fixed multiple of the population base rate.

Issue. Interpretable and defensible in principle — the operating point corresponds to a stated risk appetite (“flag borrowers at least twice as likely to default as average”). Empirically it did not outperform the ROC formulation, and it introduces an additional fitted component between the model and the decision, with its own estimation error.

Revised. Superseded by Formulation III. The isotonic machinery was retained for the expected recovery mapping, where a monotone calibrated transform is required.

Formulation III: constrained ROC sweep (retained)

The retained design abandons any fixed rule and instead searches directly for the operating point satisfying the stated constraints. The full ROC curve is computed on training data, and the threshold selected is the one maximising detection subject to the false-positive budget:

$$\tau^* = \arg \max_{\tau} \text{TPR}(\tau) \quad \text{subject to} \quad \text{FPR}(\tau) \leq 0.30 \quad (4)$$

The selected threshold is then applied to the test set without modification.

This formulation has three properties the earlier designs lack. It corresponds directly to the stated operating constraints rather than approximating them. It makes infeasibility explicit — if no threshold satisfies both constraints, the sweep reports this rather than silently returning the closest available point, and the required AUC for feasibility, $(1 + \text{TPR} - \text{FPR})/2$, can be compared against the achieved AUC to distinguish a threshold problem from a signal problem. And it separates selection from evaluation cleanly, since the sweep is confined to training data.

Scoring quantity

Previously. Thresholds scored against terminal simulated health at the forecast horizon.

Issue. The alert fires the first time a trajectory crosses the threshold, not at the horizon. Scoring against the endpoint therefore evaluated a different quantity than the alert logic uses, and a trajectory dipping below the threshold and recovering would be scored as though it never alerted.

Revised. Thresholds scored against minimum health reached over the horizon, matching the first-crossing semantics of the alert.

Forecast horizon

Previously. Calibration horizon of twelve months.

Issue. Trajectory separation between eventual defaulters and non-defaulters is concentrated in the early forecast periods. Over longer horizons the recursive blend compounds, and the marginal information in each additional step declines relative to accumulated forecast variance.

Revised. Calibration horizon reduced to three months. The thirty-six month simulation is retained for trajectory visualisation and monitoring tier assignment, where the longer view is informative, but threshold calibration and alert evaluation operate on the shorter horizon.

Signal selection

Several candidate signals are available for the alert decision: the simulated trajectory, the health score directly, the forecast increment, the external bureau score, and a direct classifier. Rather than fixing the choice a priori, all candidates are scored on training data and the strongest is selected, with the selection reported alongside test performance for each candidate.

Previously. Candidate signals compared on training AUC, with the direct classifier fit and scored on the same partition.

Issue. In-sample scoring necessarily advantages a fitted model over candidate signals not fit on that data. The classifier reported an AUC of 0.998 and won selection, then generalised worst of all candidates. The defect corrupted a selection decision rather than a reported metric, making it invisible in the headline results.

Revised. Direct classifier scored by out-of-fold prediction under five-fold stratified cross-validation, placing all candidates on comparable footing.

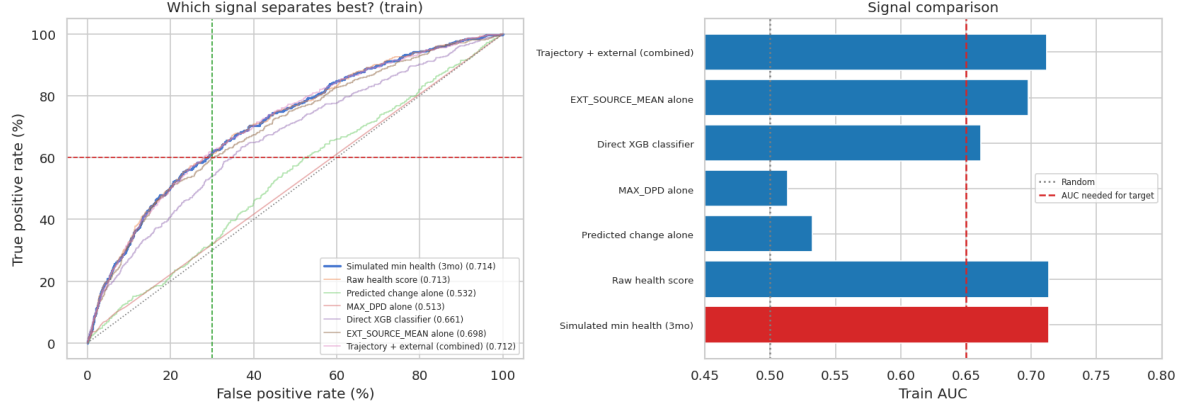


Figure 4: Candidate signal comparison on training data. Left: ROC curves for all candidates, with the selected signal emphasised; the vertical and horizontal rules mark the false-positive budget and detection target respectively. Right: AUC by candidate, against the AUC required for the target operating point to be feasible.

5 Data Integrity Audit

The initial pipeline reported $R^2 = 0.985$. A systematic audit identified seven defects, four constituting information leakage. Corrected, the model reports $R^2 = 0.32$.

Previously. Model performance accepted on the basis of held-out evaluation alone.

Issue. A train/test partition detects overfitting but not leakage. Where a feature encodes information unavailable at prediction time, that information is present in both partitions, and held-out performance is inflated rather than corrected. An implausibly strong result on a noisy behavioural forecast is therefore evidence of a construction defect, not of model quality.

Revised. Systematic audit of every feature against a single criterion: whether its value at period t could have been computed using only information available before period t . Each defect below was identified by that test.

5.1 Latent stress estimated across full borrower history

Documented in Section 4.2. An observation at month 5 carried a feature informed by months 6 onward.

5.2 Mixture models fit prior to partition

Cause. Both mixture models were fit before the train/test split, so component boundaries were informed by test-set borrowers.

Correction. Borrower-level partition performed first; models fit on training data and applied to test by transform only. Partition disjointness asserted programmatically.

5.3 Acceleration feature reconstructing its target

Cause. The feature was constructed as $\text{RAW_CHANGE}[t] - \text{VELOCITY_LAG1}[t]$, incorporating the current period. The prediction target was an exponentially-weighted mean (span 2) of that same $\text{RAW_CHANGE}[t]$, weighting it at approximately 67%. Therefore

$$\text{target}[t] \approx 0.67 \cdot \text{ACCELERATION}[t] + \text{VELOCITY_LAG1}[t] \quad (5)$$

The model was solving an algebraic identity rather than learning a relationship. The two features accounted for 55% of total feature importance.

Correction. Feature reconstructed from lagged values exclusively. Leading feature importance subsequently fell to 6.5%.

5.4 Contemporaneous features predicting same-period change

Cause. Features describing period t were the same inputs used to construct the health score at period t , rendering prediction of the change into period t circular. This additionally created a mismatch between training conditions and simulation conditions, in which only lagged information is available.

Correction. All snapshot features shifted to prior-period values.

Combined effect of Sections 5.1–5.4. R^2 reduced from 0.985 to 0.32.

5.5 Cox model misspecification

Cause. The model was fit at one observation per borrower-month, each labelled with the same eventual outcome but a different duration, treating repeated correlated measurements of a single subject as independent observations. This violates the independence assumption of the Cox partial likelihood and inflates the effective sample size.

Correction. Respecified to one observation per borrower, with durations derived from observed day-spans rather than binned approximations.

5.6 Positional rather than temporal lagging

Cause. Row-position shifts assume an observation in every consecutive period. A borrower observed at months 3, 4, 7, 8 resolved month 7's prior observation to month 4, treating a three-month transition as a single-month step.

Correction, first iteration. Explicit gap validation with velocity normalised by gap length. This discarded 84% of observations, since floor-based bucketing against a global grid misclassifies consecutive payments under real 30.44-day schedules: whether two genuinely consecutive installments fall one or two buckets apart depends on their position relative to an arbitrary boundary.

Correction, second iteration. Bucket boundaries anchored to each borrower's first installment, with rounding rather than flooring. Observation loss reduced to 56%; effective test sample increased from 633 to 1,673 borrowers.

5.7 In-sample scoring of the baseline classifier

Observation. The direct classification baseline reported train AUC 0.998 and won signal selection, then performed substantially worse on test.

Cause. The classifier was fit and scored on the same partition. In-sample scoring necessarily advantages a fitted model over candidate signals not fit on that data.

Impact. This defect did not inflate a reported metric. It silently corrupted a model selection decision, causing the least generalisable candidate to be selected.

Correction. Out-of-fold prediction under five-fold stratified cross-validation. The selection procedure is described in Section 4.8.

6 Results

6.1 Evaluation design

Previously. Alert performance and trajectory validation assessed on subsamples of 100 to 300 borrowers.

Issue. At an approximate 8% base rate, a 100-borrower subsample contains roughly eight defaulters. Detection rates estimated from such a sample carry intervals wide enough to accommodate most plausible values, and are unstable under reseeding.

Revised. Evaluation extended to the full test partition. The reported operating point is computed over 1,673 borrowers containing 138 eventual defaulters.

Previously. Model quality reported as AUC alone.

Issue. AUC summarises ranking across all thresholds and does not indicate whether any single threshold satisfies a stated operating constraint. Two models with identical AUC can differ materially at a fixed false-positive budget.

Revised. Detection rate, false-positive rate, precision and lift reported at the selected operating point, alongside AUC with interval estimates.

6.2 Behavioural score versus commercial bureau score

Table 3: Candidate signal performance, behavioural pipeline versus external bureau score

Signal	Train AUC	Test AUC	95% CI (test)
Raw health score (behavioural)	0.7134	0.7333	[0.684, 0.782]
Trajectory + external combined	0.7119	0.7302	[0.681, 0.779]
Simulated min health (behavioural)	0.7136	0.7300	[0.681, 0.779]
Home Credit external bureau score	0.6975	0.7220	[0.672, 0.772]
Direct XGBoost (out-of-fold)	0.6614	0.7073	—
MAX_DPD alone	0.5135	0.5178	—

At a standard error of approximately 0.025 on 138 test defaulters, the behavioural signals and the bureau score are statistically indistinguishable.

This establishes the principal finding: repayment behaviour observable to a single lender carries default signal equivalent to a bureau score assembled from a borrower’s complete credit footprint. The mechanism is identifiable in the feature set — fragmentation, timing dispersion, and obligation prioritisation are transaction-level signals not available at bureau level.

Ablation. Removing EXT_SOURCE_* from the direct classifier reduces performance from 0.707 to 0.612, confirming the external scores carry substantive signal. The behavioural pipeline achieves 0.730 without dependence on them.

Comparison against unconstrained modelling. The staged pipeline outperforms a direct XGBoost classifier on identical features (0.730 versus 0.707), indicating that decomposition into scoring, forecasting, and simulation stages generalises more effectively than a single unconstrained model.

6.3 Operating point

Table 4: Test-set operating point against design targets

Metric	Achieved	Design target
Detection rate (TPR)	64.5%	$\geq 60\%$
False positive rate	30.1%	$< 30\%$
Precision	16.2%	base rate 8.2%
Lift over base rate	$1.96\times$	—
Test AUC	0.7300	train 0.7136

The operating threshold was selected by sweeping the ROC curve on training data for the constrained operating point, then applied without modification to the test set. Flagged accounts default at approximately twice the population rate.

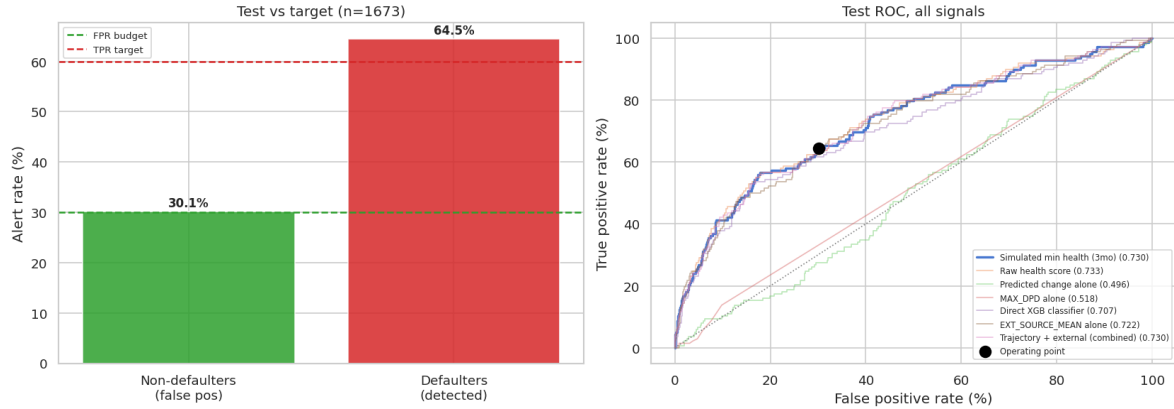


Figure 5: Test-set performance. Left: alert rate among eventual defaulters and non-defaulters against the design targets. Right: test ROC curves for all candidate signals with the selected operating point marked, overlaid on the corresponding training curve.

6.4 Survival validation

$$\text{C-index} = 0.798 \quad (6)$$

Specified as one observation per borrower with durations from observed day-spans, the model confirms the health score as a strong time-to-default predictor, with concordance approaching 0.80.

6.5 Operational output

The simulation layer produces outputs unavailable from a point-in-time score:

- **Alert timing** with quantified lead time to threshold crossing
- **Monitoring tier assignment** scaling review frequency to forecast risk

- **Expected recovery** per account, translating forecast health into an expected shortfall on the next installment

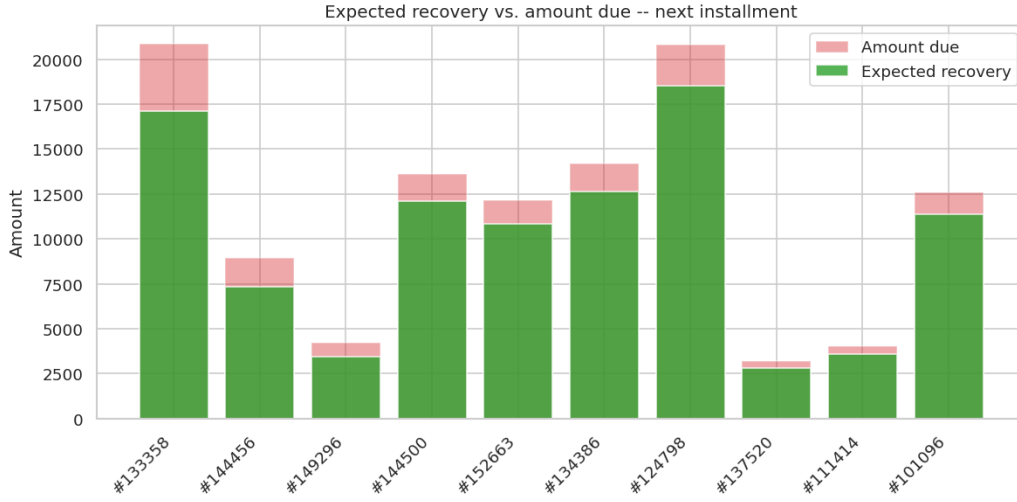


Figure 6: Expected recovery against amount due for the flagged at-risk batch. The difference between the two series is the projected shortfall on the next installment, computed prior to the payment date.

7 Conclusions

A behavioural credit score constructed exclusively from repayment records achieves parity with a commercial bureau score on held-out borrowers. The result rests on transaction-level signals — payment fragmentation, timing dispersion, and obligation prioritisation — that are structurally unavailable to bureau-level aggregation.

The staged architecture generalises more effectively than an unconstrained classifier on identical features, and the survival specification confirms the health score as a strong time-to-default predictor independently of the classification framing.

The systematic integrity audit is a substantive component of the work. Four leakage pathways and three methodological defects were identified and corrected, reducing reported R^2 from 0.985 to 0.32 and redistributing feature importance from 55% concentration in a single leaked feature to 6.5% in the leading legitimate feature.

8 Further Development

1. **Recovery of remaining observation loss.** Calendar-based rather than fixed-interval bucketing would recover a substantial portion of the 56% currently excluded by gap validation, narrowing confidence intervals across all reported results.
2. **Integration of bureau.csv and previous_application.csv.** These sources represent the standard route to higher performance on this dataset and would permit evaluation against a full-information benchmark rather than external scores alone.
3. **Extended observation windows.** Longer per-borrower histories would strengthen trajectory estimation.
4. **Cost-weighted threshold selection.** Replacing the constrained operating point with an explicit cost function over false positives and missed defaults would align the operating point with realised collections economics.

9 References

- [1] Home Credit Group (2018). *Home Credit Default Risk*. Kaggle Competition Dataset. <https://www.kaggle.com/c/home-credit-default-risk>
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- [3] Cox, D. R. (1972). Regression Models and Life-Tables. *Journal of the Royal Statistical Society: Series B*, 34(2), 187–202.
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(Provides the sampling framework underlying the latent stress estimation in Section 4.2.)
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